Certified Access: Test-Time Compute on Conservative Physics-Structured AI Memory

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Each test-time compute mechanism carries a certificate.

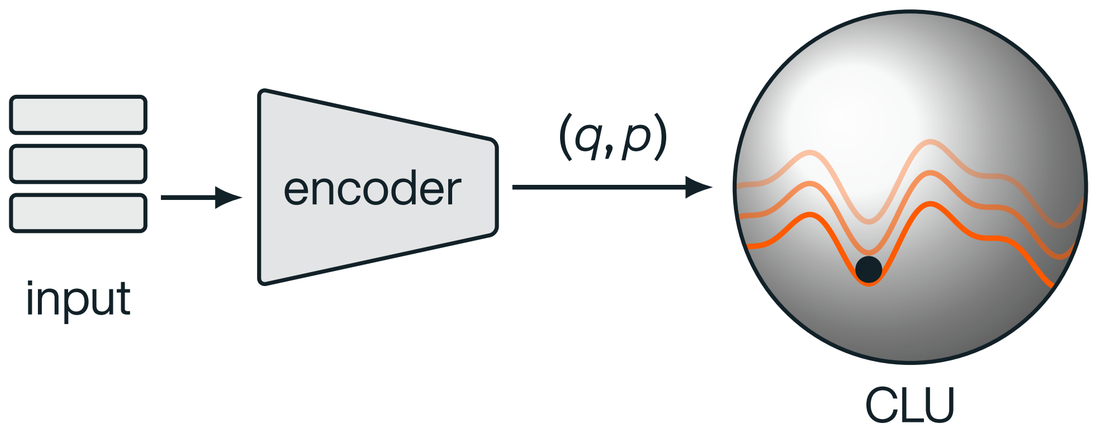
Verified on a designed test system. The guarantees hold only under stated assumptions.

1
det J: the gated channel keeps volume

0
det J: the replacing map destroys it

4.81×
fewer steps, memory of 16 pairs

1 Compute is access to phase space.



$$H(q,p) = T(p) + V_\theta(q), \quad T(p) = \sqrt{c^2\, p^\top M^{-1} p + m_0^2 c^4}$$

- Terms: c speed limit, M learned mass, m_0 rest mass, V_θ learned potential, ε step size.
- Causal box: the kinetic term caps each speed at $c/\sqrt{M_i}$, so in T steps the state stays within $L_i = T\varepsilon c/\sqrt{M_i}$ of its start.
- Two failures: stuck behind a barrier (escape), or too far away (reach).

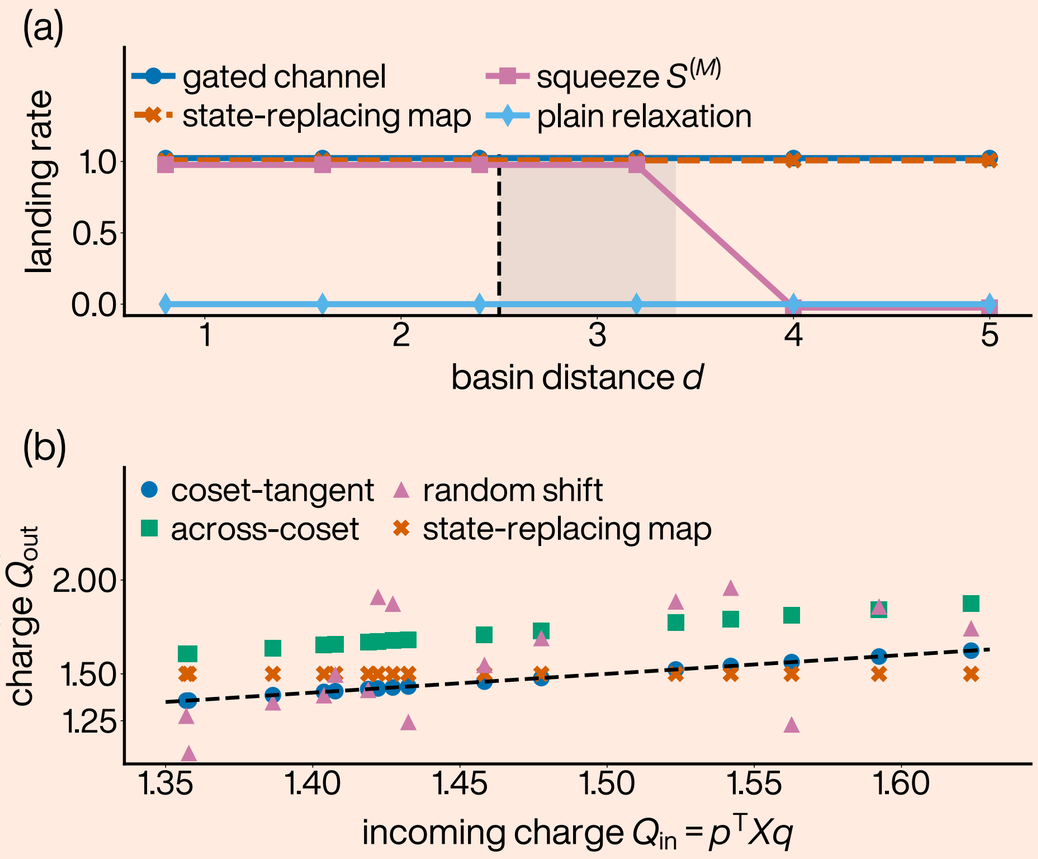
2 Access has a certificate and a cost.

- Volume: how much phase space a step keeps (det J).
- Energy: how much a step adds, bounded.
- Latch transport: how stored content moves.
- Squeeze cost: energy at most $e^{2|\zeta|}H$. Reaching $d = 4.0$ needs $\zeta \approx 2.01$, and $d = 5.0$ needs $\zeta \approx 2.64$.
- Channel cost: the potential difference, zero here.

3 The guarantees have limits.

- They hold only under the stated assumptions, on a designed test system.
- They do not cover the state-dependent damping.
- The latch law holds at the jump only.
- A jump that costs no energy can still escape.
- We do not show that the full routing policy can be undone.

4 A channel transports content; a state-replacing map erases it.



Designed test system. (b) Dashed: identity. The det $J = 0$ map sends all 16 states to one charge.

5 A learned gate rations compute.

Memory size	16 pairs	24 pairs	32 pairs
Fewer steps	4.81±0.44×	1.57±0.07×	1.14±0.06×

- Gains shrink with memory size; at full training the gate only matches the full budget.
- Routing: a router without physics does better.

6 Access comes with a certificate.

- Result: the channel reaches at det $J = 1$; the squeeze resolves escape; the gate rations compute.
- Next: read-budget trade-off (appendix).

40th Conference on Neural Information Processing Systems (NeurIPS 2026).
Workshop: Long-Context Foundation Models.

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